

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Date: 12 June 2026	Maximum Marks: 4 Marks
Team ID:	Project Name: E-commerce Application (ShopEZ)

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	• Registration through Form • Registration through Gmail • Registration through LinkedIN
FR-2	User Confirmation	• Confirmation via Email • Confirmation via OTP
FR-3	Product Browsing and Search	• Browse products by categories and sub-categories • Keyword-based search with predictive text filters • Sort and filter by price, rating, brand, and availability
FR-4	Cart and Checkout Management	• Add, update, and remove items from the shopping cart • Apply promotional discount codes and coupons • Multi-step secure checkout with order summary breakdown
FR-5	Payment Gateway Integration	• Process payments via Credit/Debit Cards • Support Net Banking and Unified Payments Interface (UPI) • Generate automated digital invoices and transaction logs
FR-6	Order Tracking and History	• View historical order records and order details • Real-time order status tracking (Dispatched, In-Transit, Delivered)
FR-7	Customer Reviews and Ratings	• Submit star ratings and text reviews for purchased items • Upload product images within reviews
FR-8	Admin Product Management	• Add, edit, and delete product listings and media • Real-time inventory tracking and low-stock alerts

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-9	Admin Order and Sales Analytics	• Manage order fulfillment status and processing pipeline • View dashboard analytics for sales revenue and top products

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The user interface must be fully responsive across mobile, tablet, and desktop viewports, featuring an intuitive layout that requires no formal user training.
NFR-2	Security	All user passwords must be hashed using bcrypt, and sensitive data transactions must be encrypted in transit using SSL/TLS protocols.
NFR-3	Reliability	The platform must maintain a database transactional consistency rate of 99.99% and ensure zero data loss during concurrent checkout sessions.
NFR-4	Performance	The system must ensure that the homepage and product search results render completely in less than 2 seconds under normal network conditions.
NFR-5	Availability	The web application must achieve an operational uptime objective of 99.9% annually, excluding scheduled system maintenance windows.
NFR-6	Scalability	The application architecture must scale horizontally to handle up to 5,000 concurrent active users during peak shopping traffic hours without degradation.